

Green Technology Transition - Game

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At the outset, we consider a market with N players. Each player $i \in \{1, 2, \dots, N\}$ is assigned a market share $s_i \in (0, 1)$, such that $\sum_{i=1}^N s_i = 1$.

Each player chooses an investment level in green technology, where the feasible set is $a_i \in [0, s_i]$.

Green demand is uncertain and represented by the random variable G . Its realized value is denoted by g . In the case of deterministic demand, we simply write g .

We assume that green customers are initially allocated across firms proportionally to their market shares. Thus, firm i is initially matched with gs_i green customers.

The relationship between investment a_i and allocated demand gs_i determines market dynamics, where the game element comes in:

- If $a_i < gs_i$, firm i may fail to serve all its potential green customers and can lose up to $gs_i - a_i$ customers.
- If $a_i = gs_i$, the firm fully serves its allocated demand, neither losing customers nor attracting additional ones.
- If $a_i > gs_i$, the firm has excess green capacity and may attract customers from competitors, gaining up to $a_i - gs_i$.

The position $a_i - gs_i$ denotes whether the firm i may lose green customers (i.e., < 0) or gain green customers (i.e., > 0). The group of players with positive positions, denoted as N^+ , may steal from the group of players with negative positions, denoted as N^- . Players in N^+ is indexed by z , and Players in N^- is

	#1	#2	#3
Market share s_i	0.2	0.3	0.5
Action a_i	0.0	0.2	0.45
Green market share gs_i	0.1	0.15	0.25
Position $a_i - gs_i$	-0.1	0.05	0.2
New market share s_i^{new}	0.1	0.3375	0.5625

Table 1: Green demand $G \sim U(0.1, 0.9)$ with realized $g = 0.5$, with $\beta = 0$.

indexed by j . The stealing rule is as follows: the positive-positioned player gains green market shares by the proportion of $(1 - \beta) \cdot s_z + \beta \cdot \log(1 + (a_z - gs_z))$, while the negative-positioned player loses green market share by the proportion of $(1 - \alpha) \cdot s_j + \alpha \cdot \log(1 + (gs_j - a_j))$.

The parameters β and α determine how market shares are redistributed. When $\beta = 1$, the split depends entirely on green investment efforts, whereas when $\beta = 0$, the split depends solely on initial market shares. The same intuition applies to α .

The second component is the reward r_i . The trade-off (also the learning point for students) is that a firm might invest a lot in green technology capacity and gain market share, but its actual reward is low. There is a sweet spot that balances market expansion and reward measured in profits.